

Independent events

A and B are independent when knowing B does not change the probability of A, provided $P(B) > 0$.

Would the first toss of a fair coin change the chance of heads on a separate independent toss?

Test independence

$P(A) = 0.4$, $P(B) = 0.5$, and $P(A \text{ AND } B) = 0.2$.

$$P(A \cap B) = P(A)P(B)$$

Are A and B independent?

Dependence

Dependent events are events for which knowing one occurred changes the probability of the other. $P(A) = 0.4$; $P(A | B) = 0.7$.

What does knowing B change?

Independent product

Two printers operate independently. Their probabilities of completing a job without a jam are 0.9 and 0.8.

What is the probability both complete without a jam?

Two successes

Two independent fair-coin tosses are made.

What is the probability of heads on both tosses?

Two outcomes in one roll

A fair die is rolled. A = even; B = greater than 3.

Are A and B independent?

Counting a combined sample space

Flip a fair coin and independently roll a fair six-sided die.

How would you list all outcomes and find $P(\text{heads AND an even number})$?

Mutually exclusive events

Mutually exclusive, or disjoint, events share no outcomes and cannot occur together.

On one die roll, are “a 2” and “a 5” mutually exclusive?

Not mutually exclusive

One card is drawn from a standard deck. A = heart; B = king.

Can both events happen together?

Positive-probability disjoint events

A and B are mutually exclusive, with $P(A) = 0.3$ and $P(B) = 0.4$.

If B occurs, what is $P(A | B)$?

Disjoint or independent?

A and B are mutually exclusive and both have positive probability.

Can A and B also be independent?

With replacement

A box contains 3 red and 2 blue tokens. Draw, replace, mix, and draw again uniformly.

Are the two red-draw events independent?

Without replacement

A box contains 3 red and 2 blue tokens. Draw two uniformly without replacement.

What is the probability of a red second draw given a red first draw?

Which assumption fails?

A student answers a series of questions while receiving coaching after every response.

Why might a constant-probability independent-trial model be questionable?

A conditional check

$P(A) = 0.25$ and $P(A | B) = 0.25$, with $P(B) > 0$.

What does this tell us about independence?

Complementary events

A has probability 0.6 and A-complement has probability 0.4.

Are these events mutually exclusive, independent, or both?

An independence model

Two machines share the same power supply.

What might make their failure events dependent?

Empirical association

Synthetic survey: 60% of all students use laptops, but 80% of tablet users do.

Are laptop use and tablet use independent in these data?

Coin tosses and “due” outcomes

An independent fair coin has landed tails six times in a row.

What is the probability of heads on the next toss?

Trials versus events

A single die roll has events $A = \text{even}$ and $B = \text{odd}$.

Why does using a fair die not make these two events independent?

Prepare a repeated-trial model

A binomial model requires independent trials with the same success probability.

Which is closer: draws with replacement or draws without replacement from a small box?